Algorithms and Mathematical Concepts for Computing

11th November 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

Consider the following scenario:

You are tasked with developing a software application for a library to manage their book lending system. The application needs to track book availability, update lending statuses and generate reports on lending trends.

a) State in which stage of the software development process user interface layouts are created, based on the initial requirements and defined project scope. **1**

Mark scheme
Design (1 mark)

b) State which stage of the software development process involves locating bugs and ensuring it meets the specified requirements before deployment. **1**

Mark scheme
Testing (1 mark)

© NCC Education Limited 2024

		Marks
c)	Explain the purpose and benefits of encapsulation and modularity in software design. Give ONE (1) real-life example for each encapsulation and modularity.	**8**

Mark scheme
Criteria:
Encapsulation is a programming concept that* restricts access to certain components of an object and only exposes the necessary parts. *(1 mark)
This improves code maintenance by hiding the internal state and requiring all interaction to occur through well-defined interfaces. (1 mark)

Benefits: (Maximum of 1 mark for writing any of the benefits listed below.)
- Protects the integrity of the data.
- Reduces complexity.

Example: A real-life example is a remote control for a TV. The user interacts with buttons without knowing the internal workings of the TV. (1 mark. Any other valid real-life example will also be accepted.)

Modularity refers to the design principle of breaking down a program into separate components or modules (1 mark) that can be developed, tested and maintained independently. (1 mark)

Benefits: (Maximum of 1 mark for writing any of the benefits listed below.)

- Easier to manage and update.
- Promotes code reuse.

Example: A real-life example is a car, which consists of various modules (engine, wheels, brakes) that can be independently designed and repaired. (1 mark. Any other valid real-life example will also be accepted.)

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 2

Consider the following content related to numbers and number systems:

a) Which of the following is **NOT** a type of number? **1**

i) Integer
ii) Fraction
iii) Decimal
iv) Symbol

Mark scheme
iv) Symbol (1 mark)

b) Fill in the missing part of the following statement: **1**
Decimal to _____ conversion involves converting a number from base 10 to base 16.

Mark scheme
Hexadecimal (1 mark)

c) Describe the concept of greatest common divisor (GCD) **and** explain its significance in number theory. **2**

Mark scheme

- ***The Greatest Common Divisor (GCD) is the largest positive integer that divides all the numbers in a given set without leaving a remainder. (1 mark)***
- ***It is significant in number theory for simplifying fractions and solving problems involving divisibility.***
  (1 mark)

d) Perform the following conversions and show your working. **2**

i) Convert the octal number **76** to binary number.

Mark scheme:
- 7 in octal = 111 in binary

- 6 in octal = 110 in binary

So, 76 (octal) = 111110 (binary).

(1 mark for correct answer, 1 mark for working)

© NCC Education Limited 2024

Marks

ii) Convert the hexadecimal number 2F to decimal number. **2**

Mark scheme:

2F = (2 × 16^1) + (15 × 16^0)

=> 32 + 15

=> 47 (decimal).

(1 mark for correct answer, 1 mark for working)

iii) Convert the decimal number **100** to binary number. **2**

Mark scheme:
Convert 100 to binary:

1. ***100 ÷ 2 = 50, remainder 0***
2. ***50 ÷ 2 = 25, remainder 0***
3. ***25 ÷ 2 = 12, remainder 1***
4. ***12 ÷ 2 = 6, remainder 0***
5. ***6 ÷ 2 = 3, remainder 0***
6. ***3 ÷ 2 = 1, remainder 1***
7. ***1 ÷ 2 = 0, remainder 1***

Now, write the remainders from bottom to top:
1100100
So, the binary equivalent of 100 is 1100100_2.

(1 mark for correct answer, 1 mark for working)

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following operators represents the logical NOT operation? **1**

i) $\wedge$
ii) $\vee$
iii) !
iv) $\rightarrow$

Mark scheme

iii) ! ***(1 mark)***

b) Complete the truth table for the Boolean expression: **6**

$$(P \wedge \neg Q) \vee (Q)$$

P	**Q**	$\neg Q$	$P \wedge \neg Q$	$(P \wedge \neg Q) \vee (Q)$
T	T			
T	F			
F	T			
F	F			

Mark scheme
Accurate completion of the truth table (Each correct column = 2 marks. If 2 out of 4 values in a column are correct, 1 mark will be awarded.)

Answer:

P	**Q**	$\neg Q$	$P \wedge \neg Q$	$(P \wedge \neg Q) \vee (Q)$
T	T	F	F	T
T	F	T	T	T
F	T	F	F	T
F	F	T	F	F

c) Explain the concept of biconditional statements in Boolean algebra **and** provide **ONE (1)** example. **3**

Mark scheme
Criteria:

A biconditional statement in Boolean algebra is a statement that is true if both components are true or both are false, (1 mark) typically expressed as "P if and only if Q" (P $\leftrightarrow$ Q). (1 mark)

Example: "You can go out if and only if you finish your homework." (1 mark, other valid examples are also acceptable.)

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 4

Answer the following questions related to Predicate Logic, Set Theory and Venn Diagrams:

a) Which of the following symbols represents the subset relationship in set theory? **1**

i) ⊆
ii) ⊂
iii) ∪
iv) ∩

Mark scheme

- ***Answer: i)*** ⊆ and ***ii)*** ⊂ ***(1 mark)***

b) Explain the concept of set complement **and** provide **ONE (1)** example. **2**

Mark scheme

The concept of set complement refers to the set of all elements not in the given set, relative to a universal set. (1 mark)

For example, (1 mark)

If the universal set is U = {1, 2, 3, 4, 5} and set A = {1, 2}, then the complement of A (denoted A') is {3, 4, 5}.

c) Calculate the union of the following sets: **1**

A = {2, 4, 6} and B = {1, 2, 3}.

Mark scheme

A ∪ B = {1, 2, 3, 4, 6} (1 mark)

d) What is the Cartesian product of two sets and how is it represented? Calculate the Cartesian product of the following sets: A = {a, b} and B = {1, 2}. **4**

Mark scheme

• The Cartesian product of two sets A and B is a set of all ordered pairs (a, b) (1 mark)
• where a ∈ A and b ∈ B. (1 mark)
• For A = {a, b} and B = {1, 2}, the Cartesian product A × B = {(a, 1), (a, 2), (b, 1), (b, 2)}. (2 marks. If 2 out of 4 values in the product set are correct, 1 mark will be awarded.)

© NCC Education Limited 2024

Marks

e) Calculate the intersection of the following sets and represent your answer using the intersection symbol (∩): **2**

A = {5, 10, 15} and B = {10, 20, 30}.

Mark scheme

The intersection of sets A = {5, 10, 15} and B = {10, 20, 30} is A ∩ B = {10}. (1 mark for A ∩ B and 1 mark for the correct answer {10})

Total 10 Marks

Question 5

Answer the following questions related to statistics:

a) Discuss the concept of standard deviation as a measure of variability. Provide **one** (1) example that is appropriate and **ONE** (1) example that is inappropriate for using standard deviation. **3**

Mark scheme

Standard deviation measures how data points differ from the mean. A low value indicates data are close to the mean, while a high value shows more spread. (1 mark)

Appropriate and Inappropriate Situations

Appropriate:

- ***Useful for comparing test scores. (1 mark, any other valid Appropriate are accepted)***

Inappropriate:

- ***Not applicable to categorical data. (1 mark, any other valid Inappropriate are accepted)***

b) Fill in the missing part of the following statement: **1**

In statistics, _____ is a measure that quantifies the variability of a data set by identifying the difference between the smallest and largest values.

Mark scheme

Range (1 mark)

© NCC Education Limited 2024

Marks

c) Calculate the range of the following data set: 12, 7, 15, 10, 5. Show your working. **2**

Mark scheme

Identify the maximum and minimum values: (1 mark)

- ***Maximum value (max) = 15***
- ***Minimum value (min) = 5***

Use the formula for range: (1 mark)

- ***Range=Maximum−Minimum***
- ***Range=15−5=10***

d) Explain the concept of skewness and why it is useful. **2**

Mark scheme

- ***Skewness in statistics measures the asymmetry of a data distribution relative to its mean. (1 mark)***
- ***Skewness helps understand data distribution, affecting statistical analysis and test selection. (1 mark)***

e) Calculate the mode of the following data set: 3, 6, 3, 8, 9, 3, 10. Show your working. **2**

Mark scheme

Criteria:

- ***Count the frequency of each number: (1 mark)***
  - ***3 appears 3 times.***
  - ***6 appears 1 time.***
  - ***8 appears 1 time.***
  - ***9 appears 1 time.***
  - ***10 appears 1time.***
- ***Identify the number with the highest frequency: (1 mark)***
  - ***The number 3 has the highest frequency of 3.***

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 6

Answer the following questions related to Data Structures:

a) Which one of the following is **NOT** a characteristic of a queue? **1**

i) First-In-First-Out (FIFO) structure.
ii) Allows random access to elements.
iii) Useful for scheduling tasks.
iv) Can dynamically grow or shrink in size.

Mark scheme

ii) Allows random access to elements. ***(1 mark)***

b) State the name of a data structure in which elements can be accessed in a random manner. **1**

Mark scheme

Array (1 mark)

c) Define the data type "stack" **and** explain its role in managing data. **2**

Mark scheme

- ***Definition: A stack is a linear data structure that follows the Last-In-First-Out (LIFO) principle, where the last element added is the first one to be removed. (1 mark)***
- ***Role in Managing Data: Stacks are used to manage data in various applications, such as function calls in programming (keeping track of active subroutines), undo mechanisms in software and evaluating expressions in algorithms. (1 mark)***

© NCC Education Limited 2024

Marks

d) Write pseudocode for a function to insert an ***element*** into a ***queue***. The ***element*** and the ***queue*** should be parameters of the function. **4**

Assume the existence of a function called ***size(queue)*** that returns the number of elements in the queue, and assume the existence of a function called ***append(queue, element)*** that adds an element to the queue.

Check that the size of the queue does not exceed MAX_SIZE (a predefined constant). If it exceeds MAX_SIZE, print an error message.

Mark scheme

```
FUNCTION insertElementIntoQueue(queue, element)  (1 mark for any
meaningful function name and parameters)
   IF size(queue) < MAX_SIZE THEN (1 mark)
      append(queue, element) (1 mark)
   ELSE
      PRINT "Error: Queue is full. Cannot insert element." (1 mark)
   END IF
END FUNCTION
```

Total 10 Marks

Question 7

Answer the following questions related to Data Structures:

a) Which data structure is typically used for implementing function calls, allowing for the storage of local variables and control flow? **1**

i) Array
ii) Queue
iii) Stack
iv) Hash Table

Mark scheme

iii) Stack (***1 mark)***

b) Fill in the missing parts of the following statement: **2**
In a queue, elements are added to the _____ and removed from the _____.

Mark scheme

- ***back (1 mark)***
- ***front (1 mark)***

© NCC Education Limited 2024

Marks

c) Consider a stack initially containing the elements 100, 200, 300. Assume 100 is at the top of the stack. Perform the following operations one after the other: **3**

i) Pop one element.
ii) Push the number 300.
iii) Pop two elements.

Show the final contents of the stack after these operations.

Show your working and draw out the stack after each operation. Clearly identify the top of the stack.

Mark scheme

Step-by-Step Operations:

1. Pop one element: Remove 100 → Stack: [200, 300] (1 mark)
2. Push the number 300: Stack: [300, 200, 300] (1 mark)
3. Pop two elements: Remove 300 and then 200 → Final Stack: [300] (1 mark)

d) Provide **ONE (1)** real-world example and justification of how each queue data structure and stack data structure would be used. **4**

Mark scheme

Queue Example:

Real-World Example: Customer Service Line (1 mark)

Justification: Customers are served in the order they arrive, following the First-In-First-Out (FIFO) principle. This ensures fairness and efficient service. (1 mark)

Stack Example:

Real-World Example: Undo Functionality in Software Applications (1 mark)

Justification: The most recent action is undone first, following the Last-In-First-Out (LIFO) principle, allowing users to quickly correct mistakes. (1 mark)

Accept valid alternatives.

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following best describes time complexity in algorithm analysis? **1**

i) The minimum amount of time an algorithm requires to complete.
ii) The measure of the algorithm's runtime as the input size grows.
iii) The average-case scenario of an algorithm's efficiency.
iv) The number of operations performed by an algorithm for a given input size.

Mark scheme

ii) The measure of the algorithm's runtime as the input size grows. (1 mark)

b) Fill in the missing part of the following statement: **1**
The best case time complexity of binary search algorithms is _____.

Mark scheme

O(1) (1 mark)

© NCC Education Limited 2024

Marks

c) Write pseudocode for a function that implements a binary search algorithm to find a target value in a sorted array. The function takes the target value and array as parameters. It returns the index of the target value or -1 if the target is not found. **6**

You should assume the existence of a function called ***length(arr)*** which returns the number of elements in the array ***arr***.

Mark scheme

```
Function binarySearch(arr, target):
    low = 0
    high = length(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1

    return -1
End Function
```

Criteria:

- ***1 mark for initialisation of low and high.***
- ***1 mark for correct for loop.***
- ***1 mark for correct if statement.***
- ***1 mark for returning the target value mid.***
- ***1 mark for correct elif and else statement.***
- ***1 mark for assigning low and high by mid + 1 and mid -1 respectively***

d) Explain why a hash table can be more efficient than a binary search tree for lookups. **2**

Mark scheme

- ***Hash tables provide average O(1) time complexity for lookups, while binary search trees have O(log n) time complexity. (1 mark)***
- ***Hash tables use a hash function to compute an index for storing data, which allows for direct access, whereas binary search trees require traversal. (1 mark)***

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which sorting algorithm repeatedly compares adjacent elements and swaps them if they are in the wrong order? **1**

i) Merge Sort
ii) Bubble Sort
iii) Insertion Sort
iv) Quick Sort

Mark scheme

ii) Bubble Sort (***1 mark)***

b) State what is meant by backtracking in programming. **2**

Mark scheme

Backtracking in programming refers to an algorithmic technique for solving problems incrementally by trying partial solutions (1 mark) and then abandoning them if they are not valid, effectively retracing steps to find a solution. (1 mark)

© NCC Education Limited 2024

Marks

c) Fill in the blanks in the pseudocode for the factorial function implementing recursion: **5**

```
Function factorial(n):
   if ______ == 0:
      return ______
   return ______ * ______  (______)
End function
```

Mark scheme

```
Function factorial(n):
   if n == 0:
      return 1
   return n * factorial(n - 1)
End function
```

Criteria:

- ***First Blank:*** ***(1 mark)***
  - ***Answer: n*** ***(1 mark)***
- ***Second Blank):***
  - ***Answer: 1*** ***(1 mark)***
- ***Third Blank:***
  - ***Answer: n*** ***(1 mark)***
- ***Fourth Blank:***
  - ***Answer: factorial*** ***(1 marks)***
- ***Fifth Blank:***
  - ***Answer: n - 1*** ***(1 marks)***

d) Explain how recursion can be applied to traversing tree structures. **2**

Mark scheme

- ***Recursion can be applied to traversing tree structures by using recursive functions to visit nodes, which allows for elegant and concise tree traversal algorithms (like pre-order, in-order and post-order traversals). (1 mark)***
- ***Each function call processes one node and invokes itself on the child nodes. (1 mark)***

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 10

Answer the following questions related to tree data structures:

a) Which of the following best describes a binary tree? **1**

i) A tree structure where each node has at most two children.
ii) A tree structure where every node has exactly two children.
iii) A tree structure that stores elements in a linear fashion.
iv) A hierarchical structure where nodes are connected in a circular manner.

Mark scheme

i) A tree structure where each node has at most two children. (1 mark)

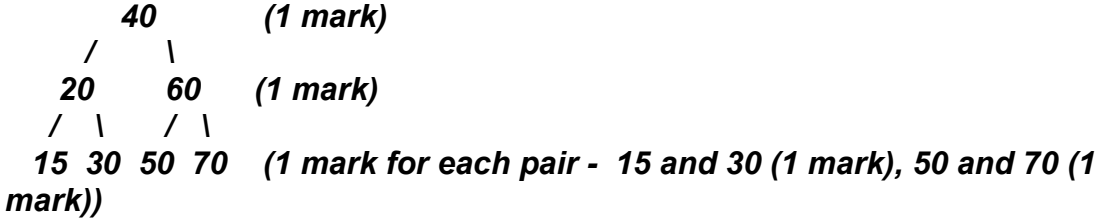
b) Fill in the missing part of the following statement: **1**
In a binary tree, a leaf node is defined as a node with _____ children.

Mark scheme

0 (1 mark)

c) Construct a binary search tree from the following sequence of numbers: 40, 20, 60, 15, 30, 50, 70. Ensure that 40 is the root of the tree. **4**

Mark scheme

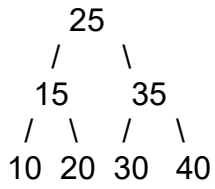


(1 mark)
(1 mark)
(1 mark for each pair - 15 and 30 (1 mark), 50 and 70 (1 mark))

© NCC Education Limited 2024

Marks

d) Using the provided binary search tree (BST) below, demonstrate the steps involved in searching for the value 30. Begin the search at the root node and describe each comparison and movement until the value is found or determined to be absent. **4**



Mark scheme

- ***1. Start at the root node (25):***
  - 30 > 25, move to the right subtree (35). (2 marks: 1 mark for checking if 30 > 25, and 1 mark for moving to the right subtree.)
- ***2. Check node (35):***
  - 30 < 35, move to the left subtree (30). Value found! (2 marks: 1 mark for checking if 30 < 35, and 1 mark for moving to the left subtree.)

Total 10 Marks

End of paper

© NCC Education Limited 2024

Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO1	Yes
2	LO5	Yes
3	LO2, LO3	Yes
4	LO4, LO6	Yes
5	LO9	Yes
6	LO7	Yes
7	LO7	Yes
8	LO8	Yes
9	LO8	Yes
10	LO7, LO8	Yes

Grade descriptors.

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Explain and use Computational Thinking	Demonstrates no understanding of Computational Thinking and its role.	Provides a basic explanation of Computational Thinking.	Discusses specific examples of Computational Thinking in various contexts.	Applies Computational Thinking to solve complex problems effectively.	Provides critical analysis and innovative approaches to Computational Thinking.
Explain and apply Boolean algebra concepts and notation.	Unable to explain Boolean algebra concepts and notation.	Provides a basic explanation of Boolean algebra concepts and notation.	Draws logic diagrams and writes Boolean expressions accurately.	Applies Boolean algebra concepts proficiently in various computing scenarios.	Offers critical analysis and optimizations of Boolean algebra operations.
Explain and use propositional logic concepts and notation	Demonstrates no understanding of propositional logic concepts and notation.	Provides a basic explanation of propositional logic concepts and notation.	Uses truth tables proficiently to evaluate propositions.	Applies propositional logic effectively in analysing complex statements.	Offers critical evaluation of propositional logic limitations and implications.
Explain and use simple predicate logic statements	Unable to explain simple predicate logic statements.	Provides a basic explanation of simple predicate logic statements.	Writes and interprets predicate logic statements involving quantifiers accurately.	Applies predicate logic statements effectively in real-world scenarios.	Provides innovative applications and thorough analysis of predicate logic statements.
Understand and explain number based system	Unable to understand or explain number-based systems.	Provides a basic explanation of number-based systems.	Performs conversions and arithmetic operations accurately in different number bases.	Applies number-based systems proficiently in solving computational problems.	Offers advanced insights and applications of number-based systems.

© NCC Education Limited 2024

Explain and use set theory and notation	Unable to explain set theory concepts and notation.	Provides a basic explanation of set theory and notation.	Uses set notation proficiently to perform operations and interpret Venn diagrams.	Applies set theory concepts effectively in analysing complex relationships.	Offers advanced interpretations and applications of set theory concepts.
Identify and use common data structures	Demonstrates no understanding of common data structures.	Provides a basic explanation of common data structures.	Applies common data structures proficiently in solving computational problems.	Demonstrates advanced understanding and optimization of data structure usage.	Offers innovative applications and optimizations of common data structures.
Explain and use common algorithms	Unable to explain common algorithms for searching & sorting.	Provides a basic explanation of common algorithms for searching & sorting.	Explains efficiency criteria used to evaluate algorithms effectively.	Applies common algorithms proficiently in solving computational problems.	Offers critical analysis and optimizations of algorithm efficiency.
Understand the fundamental concepts on statistics	Unable to understand or explain key statistical measures and concepts.	Provides a basic explanation of key statistical measures and concepts.	Calculates and uses descriptive statistics effectively.	Applies statistical concepts proficiently in analysing real-world data.	Offers advanced insights and applications of statistical concepts

© NCC Education Limited 2024